

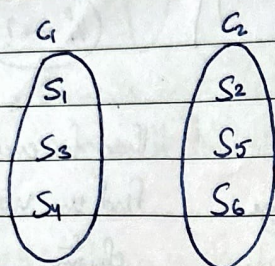
Numerical:

	x_1	x_2
S_1	1	1
S_2	1	2
S_3	2	2
S_4	8	8
S_5	8	9
S_6	9	8

 $k=2$ $k=1$ $k=2$

x_1	x_2	x_1	x_2
1	1	1	2
2	2	8	9
8	8		

Step 1



Randomly assign cluster to observations

Step 2: Compute centroids for all the clusters i.e. C_1, C_2

$$S_1: (1, 1), S_3: (2, 2), S_4: (8, 8) = \frac{1+2+8}{3}, \frac{1+2+8}{3} = (11/3, 11/3) \rightarrow C_1$$

$$S_2: (1, 2), S_5: (8, 9), S_6: (9, 8) = \frac{1+8+9}{3}, \frac{2+9+8}{3} = (18/3, 19/3) \rightarrow C_2$$

$$(6, 19/3) \rightarrow C_2$$

Step 3: Process each point and assign the cluster based on the min distance to the centroid.

$$\text{dist} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$S_1: \sqrt{\left(\frac{11}{3} - 1\right)^2 + \left(\frac{11}{3} - 1\right)^2} = 3.76$$

$$S_1: \sqrt{(6 - 1)^2 + (19/3 - 1)^2} = 7.31$$

$$S_2: \sqrt{\left(\frac{11}{3} - 1\right)^2 + \left(\frac{11}{3} - 2\right)^2} = 3.1$$

$$S_2: \sqrt{(6 - 1)^2 + (19/3 - 2)^2} = 6.61$$

$$S_3: \sqrt{\left(\frac{11}{3} - 2\right)^2 + \left(\frac{11}{3} - 2\right)^2} = 2.3$$

$$S_3: \sqrt{(6 - 2)^2 + (19/3 - 2)^2} = 5.8$$

$$S_4: \sqrt{\left(\frac{11}{3} - 8\right)^2 + \left(\frac{11}{3} - 8\right)^2} = 6.1$$

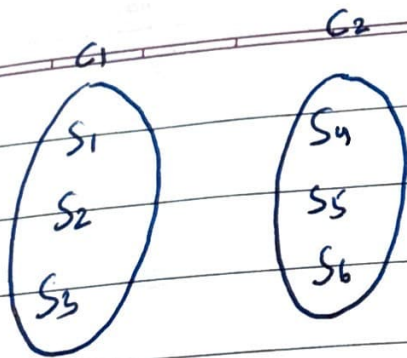
$$S_4: \sqrt{(6 - 8)^2 + (19/3 - 8)^2} = 2.5$$

$$S_5: \sqrt{\left(\frac{11}{3} - 8\right)^2 + \left(\frac{11}{3} - 9\right)^2} = 6.8$$

$$S_5: \sqrt{(6 - 8)^2 + (19/3 - 9)^2} = 3.3$$

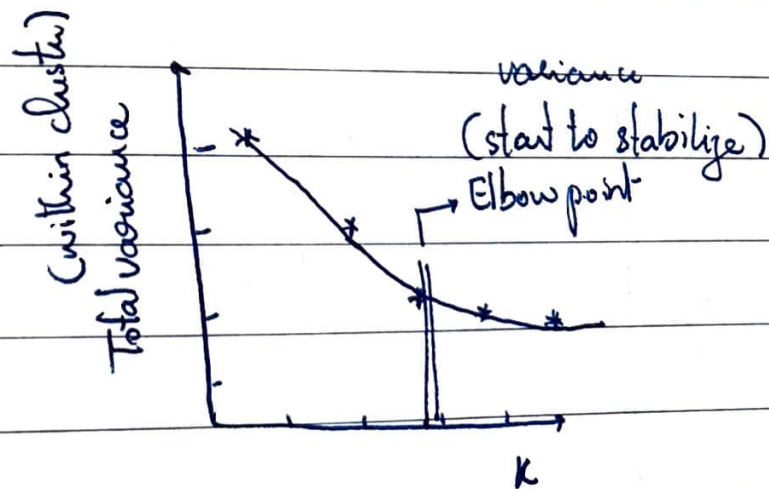
$$S_6: \sqrt{\left(\frac{11}{3} - 9\right)^2 + \left(\frac{11}{3} - 8\right)^2} = 6.8$$

$$S_6: \sqrt{(6 - 9)^2 + (19/3 - 8)^2} = 3.4$$



Iteration 1 //

Elbow Method:



- for diff values of k compute total variance within cluster
- plot the curve
- once the plateau is reached k is not needed to be increased.